

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12385505
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Bessho; Yasutaka

Centrifugal compressor

Abstract

A centrifugal compressor includes: an impeller; and a housing that accommodates the impeller, the housing including: a primary flow path that accommodates the impeller; and a secondary flow path that surrounds the primary flow path in a cross-section perpendicular to an axial direction of the impeller, the secondary flow path including: a first end that opens to the primary flow path at a position upstream of the impeller in a flow of air in the primary flow path; and a second end that opens to the primary flow path at a position downstream of the first end, the first end opening to the primary flow path in a direction opposite to the impeller in the axial direction of the impeller, the second flow path including a throttle that is continuous with the first end and that is narrowed toward the first end.

Inventors:	Bessho; Yasutaka (Tokyo, JP)
Applicant:	IHI Corporation (Tokyo, JP)
Family ID:	88419564
Assignee:	IHI Corporation (Tokyo, JP)
Appl. No.:	18/823380
Filed:	September 03, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240418185 A1	Dec. 19, 2024

Foreign Application Priority Data

JP	2022-070654	Apr. 22, 2022
----	-------------	---------------

Related U.S. Application Data

Publication Classification

Int. Cl.: **F04D29/68** (20060101); **F04D17/10** (20060101); **F04D29/42** (20060101); **F04D29/66** (20060101)

U.S. Cl.:

CPC **F04D29/685** (20130101); **F04D17/10** (20130101); **F04D29/4213** (20130101); **F04D29/66** (20130101);

Field of Classification Search

CPC: F04D (29/4213); F04D (29/685); F05D (2250/51); F05D (2220/40); F05D (2270/101)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5295785	12/1993	Church	415/58.4	F04D 29/685
5333990	12/1993	Foerster	415/58.4	F04D 29/685
7475539	12/2008	Chen	415/58.4	F04D 27/0207
8197188	12/2011	Clay	415/119	F04D 25/04
10436218	12/2018	Ozaki	N/A	F04D 29/685
2009/0263234	12/2008	Yin	N/A	N/A
2013/0302155	12/2012	Tamaki	415/203	F04D 29/441
2016/0017791	12/2015	Nakao	N/A	N/A
2018/0073515	12/2017	Kanzaka	N/A	N/A
2019/0010958	12/2018	Wang et al.	N/A	N/A
2020/0032817	12/2019	Ueno	N/A	N/A
2020/0109719	12/2019	Fujiwara et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2003-106299	12/2002	JP	N/A
2007-127108	12/2006	JP	N/A
2014-202103	12/2013	JP	N/A
2016-29273	12/2015	JP	N/A
6865604	12/2020	JP	N/A
2021-95882	12/2020	JP	N/A
WO 2016/151689	12/2015	WO	N/A
WO 2017/138199	12/2016	WO	N/A
WO 2018/198808	12/2017	WO	N/A
WO 2019/004386	12/2018	WO	N/A

OTHER PUBLICATIONS

Primary Examiner: Seabe; Justin D

Attorney, Agent or Firm: Oblon, McClelland, Maier & Neustadt, L.L.P.

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of International Application No. PCT/JP2022/048015, filed on Dec. 26, 2022, which claims priority to Japanese Patent Application No. 2022-70654 filed on Apr. 22, 2022, the entire contents of which are incorporated herein by reference.

BACKGROUND ART

Technical Field

(1) The present disclosure relates to a centrifugal compressor.

(2) In a centrifugal compressor, a secondary flow path may be formed in addition to a primary flow path (see, for example, Patent Literatures 1 to 5). A compressor impeller is arranged in the primary flow path. A first end of the secondary flow path is connected to the primary flow path at a position upstream of the compressor impeller. A second end of the secondary flow path is connected to the primary flow path at a position downstream of the first end. In low-flow rate range, a part of air pressurized by the compressor impeller may flow backward in the primary flow path. This is referred to as a surge, and limits working range of the centrifugal compressor in the low-flow rate range. However, in the centrifugal compressor with the secondary flow path, a part of the air enters the secondary flow path from the second end, and returns to the primary flow path from the first end. According to such a configuration, the effect of surge can be reduced, and the working range in the low-flow rate range can be expanded.

CITATION LIST

Patent Literature

(3) Patent Literature 1: US 2009/0263234 A Patent Literature 2: JP 6865604 B Patent Literature 3: JP 2014-202103 A Patent Literature 4: WO 2019/004386 A Patent Literature 5: JP 2021-95882 A

Technical Problem

(4) In a centrifugal compressor, it is desirable to further improve efficiency in low-flow rate range.

(5) The purpose of the present disclosure is to provide a centrifugal compressor that can improve efficiency in low-flow rate range.

Solution to Problem

(6) To solve the above problem, a centrifugal compressor according to an aspect of the present disclosure includes an impeller, and a housing that accommodates the impeller, the housing including a primary flow path that accommodates the impeller, and a secondary flow path that surrounds the primary flow path in a cross-section perpendicular to an axial direction of the impeller, the secondary flow path including a first end that opens to the primary flow path at a position upstream of the impeller in a flow of air in the primary flow path, and a second end that opens to the primary flow path at a position downstream of the first end, the first end opening to the primary flow path in a direction opposite to the impeller in the axial direction of the impeller, and the secondary flow path including an throttle that is continuous with the first end and that is narrowed toward the first end.

(7) The secondary flow path may include a plurality of fins arranged along a circumferential

direction of the impeller.

(8) The plurality of fins may be provided in the throttle of the secondary flow path.

(9) Each of the plurality of fins may be tilted with respect to a central axis of the impeller when seen in a radial direction of the impeller.

(10) Each of the plurality of fins may be tilted in the same direction as leading edges of blades of the impeller when seen in the radial direction.

Effects of Disclosure

(11) According to the present disclosure, efficiency in low-flow rate range can be improved.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematical cross-sectional view of a turbocharger including a centrifugal compressor according to an embodiment.

(2) FIG. 2 is an enlarged cross-sectional view of section Z in FIG. 1.

(3) FIG. 3 is a side view of a second insert seen in a radial direction.

(4) FIG. 4 is an enlarged cross-sectional view of a first comparative example.

(5) FIG. 5 is an enlarged cross-sectional view of a second comparative example.

(6) FIG. 6 is an enlarged cross-sectional view of a third comparative example.

(7) FIG. 7 shows results of analyses of pressure ratio in low-flow rate range.

(8) FIG. 8 shows results of analyses of efficiency in the low-flow rate range.

(9) FIG. 9 shows results of analyses of pressure ratio in high-flow rate range.

(10) FIG. 10 shows results of analyses of efficiency in the high-flow rate range.

DESCRIPTION OF EMBODIMENTS

(11) An embodiment of the present disclosure will be described in detail below with reference to the accompanying drawings. Specific dimensions, materials, and numerical values described in the embodiment are merely examples for a better understanding, and do not limit the present disclosure unless otherwise specified. In this specification and the drawings, duplicate explanations are omitted for elements having substantially the same functions and configurations by assigning the same sign. Furthermore, elements not directly related to the present disclosure are omitted from the figures.

(12) FIG. 1 is a schematical cross-sectional view of a turbocharger TC including a centrifugal compressor **100** according to an embodiment. In the present embodiment, the centrifugal compressor **100** is incorporated into the turbocharger TC. In another embodiment, the centrifugal compressor **100** may be incorporated into a device other than the turbocharger TC, or may be a stand-alone device.

(13) The turbocharger TC comprises a shaft **1**, a turbine impeller **2**, and a compressor impeller (impeller) **3**. As described later, the shaft **1**, the turbine impeller **2**, and the compressor impeller **3** rotate integrally. As such, in the present disclosure, an “axial direction,” a “radial direction,” and a “circumferential direction” of the shaft **1**, the turbine impeller **2**, and the compressor impeller **3** may simply be referred to as the “axial direction,” the “radial direction,” and the “circumferential direction,” respectively.

(14) The turbocharger TC includes a bearing housing **4**, a turbine housing **5**, and a compressor housing (housing) **6**. In the axial direction, the turbine housing **5** is connected to a first end face of the bearing housing **4** (a left end face in FIG. 1) by fastening bolts B1. In the axial direction, the compressor housing **6** is connected to a second end face of the bearing housing **4** (a right end face in FIG. 2) by fastening bolts B2.

(15) The bearing housing **4** includes a bearing hole **4**. The bearing hole **4a** extends in the axial direction in the bearing housing **4**. The bearing hole **4a** accommodates a bearing **7**. In the present

embodiment, a full floating bearing is shown as an example of the bearing 7. In another embodiment, the bearing 7 may be other radial bearing such as a semi-floating bearing or a rolling bearing. The bearing 7 rotatably supports the shaft 1.

(16) In the axial direction, a first end of the shaft 1 (a left end in FIG. 1) is provided with the turbine impeller 2. The turbine impeller 2 rotates integrally with the shaft 1. The turbine impeller 2 is rotatably accommodated in the turbine housing 5. In the axial direction, a second end of the shaft 1 (a right end in FIG. 1) is provided with the compressor impeller 3. The compressor impeller 3 rotates integrally with the shaft 1. The compressor impeller 3 is rotatably accommodated in the compressor housing 6.

(17) The compressor housing 6 includes an inlet 6a at an end opposite to the bearing housing 4 in the axial direction. The inlet 6a is connected to an air cleaner (not shown). The compressor housing 6 also includes a primary flow path 20 and a secondary flow path 30. The primary flow path 20 and secondary flow path 30 are described in detail below. The compressor impeller 3 is arranged in the primary flow path 20.

(18) The bearing housing 4 and the compressor housing 6 define a diffuser flow path 8 therebetween. The diffuser flow path 8 extends from an inner side to an outer side in the radial direction. The diffuser flow path 8 has an annular shape. The diffuser flow path 8 communicates with the inlet 6a via the compressor impeller 3.

(19) The compressor housing 6 includes a compressor scroll flow path 9. The compressor scroll flow path 9 is located at a radially outer side with respect to the diffuser flow path 8. The compressor scroll flow path 9 communicates with the diffuser flow path 8. The compressor scroll flow path 9 also communicates with an intake port of an intercooler (not shown).

(20) In the compressor housing 6 as described above, when the compressor impeller 3 rotates, air is sucked into the primary flow path 20 from the inlet 6a. The sucked air is accelerated and pressurized by centrifugal force while passing through blades of the compressor impeller 3. The air is further pressurized in the diffuser flow path 8 and the compressor scroll flow path 9. The pressurized air flows out of an outlet (not shown), and is directed to the intake port of the intercooler. In the turbocharger TC, a portion including the compressor impeller 3 and the compressor housing 6 functions as the centrifugal compressor 100.

(21) In the present disclosure, an upstream side in the flow of the sucked air in the primary flow path 20 may simply be referred to as the “upstream side,” and a downstream side in the flow of the sucked air in the primary flow path 20 may simply be referred to as the “downstream side,” with regard to the centrifugal compressor 100 and its components.

(22) The turbine housing 5 includes an outlet 5a at an end opposite to the bearing housing 4 in the axial direction. The outlet 5a is connected to an exhaust gas purifier (not shown). The turbine housing 5 includes a connecting flow path 10. The connecting flow path 10 extends from an inner side to an outer side in the radial direction. The connecting flow path 10 has an annular shape. The connecting flow path 10 communicates with the outlet 5a via the turbine impeller 2.

(23) Turbine housing 5 includes a turbine scroll flow path 11. The turbine scroll flow path 11 is located at a radially outer side with respect to the connecting flow path 10. The turbine scroll flow path 11 communicates with the connecting flow path 10. The turbine scroll flow path 11 also communicates with a gas inlet (not shown). The gas inlet receives exhaust gas discharged from an exhaust manifold of an engine (not shown).

(24) In the turbine housing 5 as described above, the exhaust gas is directed from the gas inlet to the turbine scroll flow path 11, and is further directed to the outlet 5a via the connecting flow path 10 and the turbine impeller 2. The exhaust gas rotates the turbine impeller 2 while passing through blades of the turbine impeller 2. A rotational force of the turbine impeller 2 is transmitted to the compressor impeller 3 via the shaft 1. As the compressor impeller 3 rotates, the air is sucked into the inlet 6a, and is accelerated and pressurized by the compressor impeller 3, as described above. In the turbocharger TC, a portion including the turbine impeller 2 and the turbine housing 5 functions

as a turbine **200**.

(25) Next, the primary flow path **20** and the secondary flow path **30** of the compressor housing **6** are described.

(26) FIG. **2** is an enlarged cross-sectional view of section Z in FIG. **1**. The compressor housing **6** includes a main body **60** and an insert **70**.

(27) The main body **60** includes a first cylindrical portion **61** and a second cylindrical portion **62**. Each of the first cylindrical portion **61** and the second cylindrical portion **62** has a substantially cylindrical shape.

(28) A central axis of the first cylindrical portion **61** corresponds to the central axis of the compressor impeller **3**. An inner diameter of the first cylindrical portion **61** is constant along the axial direction.

(29) The second cylindrical portion **62** is formed in and coaxially with the first cylindrical portion **61**. The first cylindrical portion **61** and the second cylindrical portion **62** are formed integrally. Upstream ends (right ends in FIG. **1**) of the first cylindrical portion **61** and the second cylindrical portion **62** are opened. Downstream ends (left ends in FIG. **1**) of the first cylindrical portion **61** and the second cylindrical portion **62** are connected to each other. In the axial direction, the second cylindrical portion **62** is shorter than the first cylindrical portion **61**, and the upstream end of the second cylindrical portion **62** is located in the first cylindrical portion **61**.

(30) The outer diameter of the second cylindrical portion **62** is constant along the axial direction, and is smaller than the inner diameter of the first cylindrical portion **61**. Accordingly, a gap is formed between the first cylindrical portion **61** and the second cylindrical portion **62** in the radial direction. An inner surface of the second cylindrical portion **62** includes a tapered portion **62a** that is tapered from the upstream side toward the downstream side.

(31) The second cylindrical portion **62** includes a slit **63**. The slit **63** penetrates the second cylindrical portion **62** in the radial direction. For example, the slit **63** may extend in the circumferential direction, and may be discontinuous in the circumferential direction. A material of this discontinuous part (rib) connects portions on both sides of the slit **63** along the axial direction. Furthermore, for example, the slit **63** may extend in the circumferential direction, and may be continuous in the circumferential direction (i.e., continuous over the entire circumference). In this case, in order to connect the first cylindrical portion **61** and the second cylindrical portion **62**, a material (rib) that is a discontinuous portion in the circumferential direction is provided in a radial gap between the first cylindrical portion **61** and the second cylindrical portion **62**.

(32) The insert **70** is arranged in the first cylindrical portion **61**. In the present embodiment, the insert **70** includes a first insert **71** and a second insert **72**.

(33) The first insert **71** has a cylindrical shape. The first insert **71** is coaxially fitted in the first cylindrical portion **61**. The first cylindrical portion **61** includes a counterbore **61a** into which the first insert **71** is fitted. In the radial direction, an inner wall of the first insert **71** is formed to be flush with an inner wall of the first cylindrical portion **61**. An inner diameter of the first insert **71** is constant along the axial direction. The first insert **71** is flush with an end face of the first cylindrical portion **61** in the axial direction, and defines the inlet **6a**. In another embodiment, the first insert **71** may be retracted or protrude from the end face of the first cylindrical portion **61** in the axial direction.

(34) The second insert **72** includes a body **73** and a plurality of fins **74**.

(35) The body **73** is coaxially arranged in the first cylindrical portion **61**. An outer surface of the body **73** includes a tapered portion **73a** that is tapered from the upstream side to the downstream side. For example, in the present embodiment, the body **73** has a substantially hollow truncated cone shape. A downstream end of the body **73** (left end in FIG. **2**) has a smaller diameter, and contacts an end of the second cylindrical portion **62**. An upstream end of the body **73** (right end in FIG. **2**) has a larger diameter, and is located in the first insert **71**. An outer diameter of the body **73** is smaller than the inner diameters of the first cylindrical portion **61** and the first insert **71**.

Accordingly, a gap is formed in the radial direction between the first cylindrical portion **61** and the first insert **71** and the body **73**.

(36) The gap between the first cylindrical portion **61** and the first insert **71** and the body **73**, the gap between the first cylindrical portion **61** and the second cylindrical portion **62**, and the slit **63** as described above serve as the secondary flow path **30**.

(37) In the first cylindrical portion **61**, a space other than the secondary flow path **30** functions as the primary flow path **20**. Specifically, a space from the inlet **6a** to the upstream end of the body **73** in the first insert **71**, a space in the second insert **72**, and a space in the second cylindrical portion **62** function as the primary flow path **20**. The primary flow path **20** extends from the inlet **6a** to a radially inner end of the diffuser flow path **8**. The secondary flow path **30** is located outside the primary flow path **20** in the radial direction in a cross-section perpendicular to the axial direction.

(38) A radial gap between the upstream end of the body **73** and the first insert **71** serves as a first end **30A** of the secondary flow path **30**. Referring to FIG. **1**, the first end **30A** opens to the primary flow path **20** at a position upstream of the compressor impeller **3**. Furthermore, in the axial direction, the first end **30A** opens to the primary flow path in a direction opposite to the compressor impeller **3**.

(39) Referring to FIG. **2**, a radially inner end of the slit **63** serves as a second end **30B** of the secondary flow path **30**. The second end **30B** opens to the primary flow path **20** at a position downstream of the first end **30A**. In the present embodiment, the second end **30B** opens to the primary flow path inwardly in the radial direction. In the present embodiment, the second end **30B** faces the compressor impeller **3** in the radial direction.

(40) The secondary flow path **30** includes a throttle **30C** that is continuous with the first end **30A** and narrowed toward the first end **30A**. Specifically, as described above, in the present embodiment, the inner diameters of the first cylindrical portion **61** and the first insert **71** are constant along the axial direction, and the outer surface of the body **73** includes a tapered portion **73a** that is tapered from the upstream side toward the downstream side. Accordingly, a space that is radially outside the tapered portion **73a** functions as the throttle **30C**.

(41) FIG. **3** is a side view of the second insert **72** seen in the radial direction. The plurality of fins **74** are arranged along the circumferential direction. The fins **74** project radially outward from the outer surface of the body **73**. The fins **74** are tilted with respect to the central axis **X** of the compressor impeller **3** when seen in the radial direction. An angle between the fin **74** and the central axis **X** is defined as α . In the present embodiment, the fins **74** are tilted in the same direction as leading edges of the compressor impeller **3** when seen in the radial direction.

(42) Referring to FIG. **2**, radially outer ends of the fins **74** contact the inner surface of the first insert **71**. The second insert **72** is fitted to the inner surface of the first insert **71** by the fins **74**. The fins **74** fix the second insert **72** to the first insert **71**.

(43) Regarding an assembly of the first insert **71** and the second insert **72**, the first insert **71** is fitted into the counterbore **61a** at first. Then, the fins **74** of the second insert **72** are fitted into the inner surface of the first insert **71**. This order allows the first insert **71** and the second insert **72** to be assembled into the main body **60**.

(44) Next, a work of the centrifugal compressor **100** is described.

(45) Firstly, a work in low-flow rate range is described.

(46) As described above, when the compressor impeller **3** rotates, air is sucked into the primary flow path **20** from the inlet **6a**. The sucked air is pressurized by centrifugal force while passing through the blades of the compressor impeller **3**.

(47) In the low-flow rate range, when a pressure at an inlet of the compressor impeller **3** is less than a pressure at an outlet of the compressor impeller **3**, a part of the pressurized air is pushed back from the outlet of the compressor impeller **3** toward the inlet of the compressor impeller **3** due to a pressure gradient being reversed. This air flows backward in the primary flow path **20**. However, since the centrifugal compressor **100** includes the secondary flow path **30**, the air flowing backward

in the primary flow path **20** enters the secondary flow path **30** from the second end **30B**. This curbs the effect of surge, and expands working range of the centrifugal compressor **100** in the low-flow rate range.

(48) The air entering the secondary flow path **30** returns to the primary flow path **20** from the first end **30A**. However, in the present embodiment, the secondary flow path includes the throttle **30C** and the fins **74** that are continuous with the first end **30A**. The throttle **30C** and the fins **74** impede the flow in the secondary flow path **30**. Accordingly, the throttle **30C** and the fins **74** can reduce the flow rate of air returning to the primary flow path **20**. As such, in the present embodiment, the secondary flow path **30** functions as a buffer region for temporarily storing air. This is the opposite of the conventional configuration where air in the secondary flow path is actively returned to the primary flow path.

(49) Furthermore, in the present embodiment, the first end **30A** opens to the primary flow path **20** in the direction opposite to the compressor impeller **3** in the axial direction. Accordingly, the secondary flow path **30** is extended in the axial direction, compared to the case where the first end **30A** opens to the primary flow path **20** in the radial direction, for example. As such, the secondary flow path **30** can store air longer.

(50) According to the above configuration, the air pressurized by the compressor impeller **3** is stored in the secondary flow path **30** for a longer time. Therefore, the air is cooled off as its pressure decreases while the air is stored in the secondary flow path **30**. As such, when returning to the primary flow path **20**, the pressure and the temperature of the air in the secondary flow path **30** approaches the pressure and the temperature of the air flowing in the primary flow path **20**. According to such a configuration, loss of flow in the primary flow path **20** can be reduced when the air in the secondary flow path **30** returns to the primary flow path **20**. As such, efficiency in the low-flow rate range can be improved.

(51) Furthermore, in the present embodiment, the fins **74** are tilted in the same direction as the leading edges of the compressor impeller **3** when seen in the radial direction. The air flowing in the primary flow path **20** is dragged by the rotation of the compressor impeller **3**, and is swirled in the same direction as the rotation of the compressor impeller **3**. In the present embodiment, since the fins **74** are tilted in the same direction as the leading edges of the compressor impeller **3**, the air returning from the first end **30A** to the primary flow path **20** is also swirled in the same direction as the rotation of the compressor impeller **3**. Accordingly, the air returning to the primary flow path **20** from the first end **30A** does not impede the flow in the primary flow path **20**. As such, loss of flow in the primary flow path **20** can be reduced when the air in the secondary flow path **30** returns to the primary flow path **20**. Thus, the efficiency in the low-flow rate range can be further improved.

(52) Next, a work in high-flow rate range is described.

(53) In the high-flow rate range, a portion of the air flowing in the primary flow path **20** enters the secondary flow path **30** from the first end **30A**, contrary to in the low-flow rate range. However, in the present embodiment, the secondary flow path **30** includes the throttle **30C** and the fins **74** that are continuous with the first end **30A**. The throttle **30C** and the fins **74** can reduce the flow rate of air entering the secondary flow path **30** from the first end **30A**. The air entering the secondary flow path **30** returns to the primary flow path **20** from the second end **30B**. In the present embodiment, a difference between a pressure of the air at the first end **30A** and a pressure of the air at the second end **30B** is relatively small, since the flow rate of the air entering the secondary flow path **30** from the first end **30A** is reduced, as described above. According to such a configuration, when the air in the secondary flow path **30** returns to the primary flow path **20** from the second end **30B**, loss of flow in the primary flow path **20** can be reduced. Accordingly, efficiency in the high-flow rate range can also be improved. Note that although the air returns to a position between the blades of the compressor impeller **3** when the air in the secondary flow path **30** returns to the primary flow path **20** from the second end **30B** in the present embodiment, the air may return to a position upstream of the compressor impeller **3**, as long as the air returns to the primary flow path **20**.

(54) Next, results of analyses of the centrifugal compressor **100** are described.

(55) FIGS. **4**, **5** and **6** are enlarged cross-sectional views of first, second, and third comparative examples, respectively.

(56) Referring to FIG. **4**, a centrifugal compressor **500** for the first comparative example differs from the above-described centrifugal compressor **100** in that the centrifugal compressor **500** does not include a secondary flow path. The centrifugal compressor **500** may be substantially similar to the centrifugal compressor **100** in other respects.

(57) Referring to FIG. **5**, a centrifugal compressor **600** for the second comparative example differs from the above-described centrifugal compressor **100** in the configuration of the secondary flow path **30D**. Specifically, the centrifugal compressor **600** does not include an insert. The secondary flow path **30D** does not include a throttle, and the first end **30A** opens wider to the primary flow path **20**, compared to that in the centrifugal compressor **100**. The centrifugal compressor **600** may be substantially similar to the centrifugal compressor **100** in other respects.

(58) Referring to FIG. **6**, a centrifugal compressor **700** for the third comparative example differs from the above-described centrifugal compressor **100** in the configuration of the secondary flow path **30E**. Specifically, in the centrifugal compressor **700**, the first end **30A** opens to the primary flow path **20** in the radial direction. Accordingly, the secondary flow path **30E** is shorter in the axial direction than the secondary flow path **30** of the above-described centrifugal compressor **100**. The insert **70** is formed by a single component. The centrifugal compressor **700** may be substantially similar to the centrifugal compressor **100** in other respects.

(59) FIGS. **7** and **8** show results of analyses of pressure ratio and efficiency in low-flow rate range, respectively. FIGS. **9** and **10** show results of analyses of pressure ratio and efficiency in high-flow rate range, respectively. In FIGS. **7** to **10**, the horizontal axis indicates the flow rate ratio. The vertical axis indicates the pressure ratio in FIGS. **7** and **9**, and the efficiency in FIGS. **8** and **10**.

(60) The “pressure ratio” in FIGS. **7** and **9** is calculated by the following equation (1), where the pressure ratio is $P_{\text{sub.}(tot)\text{ratio}}$, the pressure at the outlet of the compressor (total pressure) is $P_{\text{sub.}(tot)\text{out}}$, and the pressure at the inlet of the compressor (total pressure) is $P_{\text{sub.}(tot)\text{in}}$.

$$(61) \quad P_{\text{sub.}(tot)\text{ratio}} = P_{\text{sub.}(tot)\text{out}} / P_{\text{sub.}(tot)\text{in}} \quad (1)$$

(62) The “efficiency” in FIGS. **8** and **10** is calculated by the following equation (2), where the efficiency is $\eta_{\text{sub.}(tot)}$, the pressure ratio is $P_{\text{sub.}(tot)\text{ratio}}$, the specific heat ratio of air is $\gamma_{\text{sub.air}}$, the temperature at the outlet of the compressor (total temperature) is $T_{\text{sub.}(tot)\text{out}}$, and the temperature at the inlet of the compressor (total temperature) is $T_{\text{sub.}(tot)\text{in}}$.

$$(63) \quad \eta_{\text{sub.}(tot)} = (P_{\text{sub.}(tot)\text{ratio}} (\gamma_{\text{air}} - 1 / \gamma_{\text{air}}) - 1) / (T_{\text{sub.}(tot)\text{out}} / T_{\text{sub.}(tot)\text{in}} - 1) \quad (2)$$

(64) For example, the “flow rate ratio” in FIGS. **7**, **8**, **9** and **10** is calculated as follows. Flow rates at all rotation rates used in the analyses are calculated for each of the example and the comparative examples. Among all of the calculated flow rates including both the example and the comparative examples, the maximum flow rate at any one rotation rate is extracted. By calculating a ratio between the extracted maximum flow rate and each flow rate at the same rotation rate, the “flow rate ratio” for each of the example and the comparative examples at any one rotation rate is calculated.

(65) Properties of the centrifugal compressor **100** of the embodiment shown in FIG. **2** and the centrifugal compressors **500**, **600** and **700** of the first, second, and third comparative examples shown in FIGS. **4** to **6** were analyzed with using a fluid analysis tool. Specifically, the pressure ratios and the efficiencies were calculated for each condition when air flows at a plurality of flow rates from the inlet **6a** to the compressor impeller **3**.

(66) In FIGS. **7** to **10**, Line A-1 and Line A-2 show results of analyses of Embodiments 1 and 2 of the present disclosure, respectively, corresponding to the centrifugal compressor **100** shown in FIG. **2**. Embodiment 2 differs from Embodiment 1 in the angle α of the fin **74** and an area of the opening of the first end **30A**.

(67) Line B shows results of analyses of Comparative Example 1 corresponding to the centrifugal compressor **500** in FIG. **4** without a secondary flow path. Note that in FIGS. **7** and **8**, Line B could not be analyzed at lower flow rates compared to the other conditions.

(68) Line C shows results analyses of Comparative Example 2 corresponding to the centrifugal compressor **600** of FIG. **5**, where the first end **30A** of the secondary flow path **30D** has a wide opening to the primary flow path **20**.

(69) Lines D-**1**, D-**2** and E show results of analyses of Comparative Examples 3, 4 and 5, respectively, corresponding to the centrifugal compressor **700** of FIG. **6**, where the first end **30A** of the secondary flow path **30E** opens in the radial direction. Comparative Example 4 differs from Comparative Example 3 in the radius from the central axis X to the first end **30A**. In Comparative Examples 3 and 4, the fins **74** are oriented such that the air returned from the first end **30A** to the primary flow path **20** is parallel to the axial direction (which can also be referred to as non-swirl). In Comparative Example 5, the fins **74** are oriented such that the air returned from the secondary flow path **30** to the primary flow path **20** is swirled in a direction opposite to the rotation of the compressor impeller **3** (which can also be referred to as counter-swirl).

(70) Referring to FIG. **7**, in the low-flow rate range, the pressure ratios of Embodiments 1 and 2 (lines A-**1** and A-**2**) keep increasing as the flow rate decreases, compared to those of Comparative Examples 1 to 5 (lines B, C, D-**1**, D-**2** and E). This means that a higher-pressure ratio can be retained in the low-flow rate range, and therefore the pressure at the inlet of the compressor impeller **3**, which decreases as the flow rate decreases, can be maintained at a higher pressure. In other words, a critical flow rate range on a lower flow rate side (surge margin) can be expanded, and therefore the working range of the centrifugal compressor can be extended to a further lower flow rate side, compared to previously known structures.

(71) Referring to FIG. **8**, in the low-flow rate range, the efficiencies of Embodiments 1 and 2 (lines A-**1** and A-**2**) are higher than those of Comparative Examples 2 to 5 (lines C, D-**1**, D-**2** and E), except for Comparison Example 1 (line B). As such, according to Embodiments 1 and 2, the efficiency in the low-flow rate range can be improved.

(72) Referring to FIG. **9**, in the high-flow rate range, the pressure ratios of Embodiments 1 and 2 (lines A-**1** and A-**2**) are higher than those of Comparative Examples 1 to 5 (lines B, C, D-**1**, D-**2** and E). This means that the pressure ratio can be retained higher in the high-flow rate range, and therefore the pressure at the inlet of the compressor impeller **3**, which decreases as the flow rate increases, can be maintained at a higher pressure, with respect to an exhaust pressure resistance at a side downstream of the compressor impeller **3**. In other words, a critical flow rate range on a higher flow rate side (choke margin) can be expanded, and therefore the working range of the centrifugal compressor can be extended to a further higher flow rate side, compared to the previously known structure.

(73) Referring to FIG. **10**, in the high-flow rate range, the efficiencies of Embodiments 1 and 2 (lines A-**1** and A-**2**) are higher than those of Comparative Examples 1 to 5 (lines B, C, D-**1**, D-**2** and E). As such, according to Embodiments 1 and 2, the efficiency in the high-flow rate range can be improved.

(74) The centrifugal compressor **100** of the present embodiment as described above comprises the compressor impeller **3** and the compressor housing **6** that accommodates the compressor impeller **3**. The compressor housing **6** includes the primary flow path **20** that accommodates the compressor impeller **3** and the secondary flow path **30** that surrounds the primary flow path **20** in the cross-section perpendicular to the axial direction of the compressor impeller **3**. The secondary flow path **30** includes the first end **30A** that opens to the primary flow path **20** at the position upstream of the compressor impeller **3** in the flow of air in the primary flow path **20**, and the second end **30B** that opens to the primary flow path **20** at the position downstream of the first end **30A**. The first end **30A** opens to the primary flow path **20** in the direction opposite to the compressor impeller **3** in the axial direction, and the secondary flow path **30** includes the throttle **30C** that is continuous with the

first end **30A** and narrowed toward the first end **30A**. According to such a configuration, in the low-flow rate range, the air flowing backward in the primary flow path **20** enters the secondary flow path **30** from the second end **30B** and returns to the primary flow path **20** from the first end **30A**. However, the throttle **30C** can reduce the flow rate of the air returning from the first end **30A** to the primary flow path **20**. Accordingly, the secondary flow path **30** functions as the buffer area for temporarily storing air. Furthermore, according to the above configuration, the first end **30A** opens to the primary flow path **20** in the direction opposite to the compressor impeller **3** in the axial direction. Accordingly, the secondary flow path **30** is extended in the axial direction, as compared to the case where the first end **30A** opens to the primary flow path **20** in the radial direction, for example. As such, the secondary flow path **30** can store air for a longer time. The air pressurized by the compressor impeller **3** is cooled off as its pressure decreases while stored in the secondary flow path **30**. Accordingly, when returning to the primary flow path **20**, the pressure and the temperature of the air in the secondary flow path **30** approaches the pressure and the temperature of the air flowing in the primary flow path **20**. According to such a configuration, loss of flow in the primary flow path **20** can be reduced when the air in the secondary flow path **30** returns to the primary flow path **20** from the first end **30A**. As such, the efficiency in the low-flow rate range can be improved.

(75) In the high flow rate range, a part of the air flowing in the primary flow path **20** enters the secondary flow path **30** from the first end **30A**, contrary to in the low-flow rate range. However, the throttle **30C** can reduce the flow rate of the air entering the secondary flow path **30** from the first end **30A**. According to such a configuration, the difference between the pressure of the air at the first end **30A** and the pressure of the air at the second end **30B** is relatively smaller. Therefore, when the air in the secondary flow path **30** returns to the primary flow path **20** from the second end **30B**, loss of flow in the primary flow path **20** can be reduced. As such, the efficiency in the high-flow rate range can also be improved.

(76) Furthermore, in the centrifugal compressor **100**, the secondary flow path **30** includes the plurality of fins **74** arranged along the circumferential direction. The fins **74** can impede the flow of air in the secondary flow path **30**, allowing the air to be stored longer in the secondary flow path **30**. Accordingly, the efficiency in the low-flow rate range can be further improved.

(77) Furthermore, in the centrifugal compressor **100**, the plurality of fins **74** are provided in the throttle **30C** of the secondary flow path **30**. According to such a configuration, the fins **74** can reduce the flow rate of air entering the secondary flow path **30** from the first end **30A** in the high-flow rate range. Accordingly, the efficiency in the high-flow rate range can be further improved.

(78) Furthermore, in the centrifugal compressor **100**, each of the plurality of fins **74** is tilted with respect to the central axis **X** of the compressor impeller **3** when seen in the radial direction. According to such a configuration, the fins **74** can impede the flow of air more in the secondary flow path **30**, and can store the air in the secondary flow path **30** longer. Accordingly, the efficiency in the low-flow rate range can be further improved.

(79) Furthermore, in the centrifugal compressor **100**, each of the plurality of fins **74** is tilted in the same direction as the leading edges of the blades of the compressor impeller **3** when seen in the radial direction. According to such a configuration, the air returning to the primary flow path **20** from the first end **30A** is swirled in the same direction as the rotation of the compressor impeller **3**, i.e., the direction in which the air flowing in the primary flow path **20** is swirled. Accordingly, the air returning to the primary flow path **20** from the first end **30A** does not impede the flow in the primary flow path **20**. Therefore, loss of flow in the primary flow path **20** can be reduced when the air in the secondary flow path **30** returns to the primary flow path **20**. As such, the efficiency in the low-flow rate range can be further improved.

(80) Although the embodiment of the present disclosure has been described above with reference to the accompanying drawings, the present disclosure is not limited thereto. It is obvious that a person skilled in the art can conceive of various examples of variations or modifications within the scope of the claims, which are also understood to belong to the technical scope of the present disclosure.

Claims

1. A centrifugal compressor comprising: an impeller; and a housing that accommodates the impeller, the housing including: a primary flow path that accommodates the impeller; and a secondary flow path that surrounds the primary flow path in a cross-section perpendicular to an axial direction of the impeller, the secondary flow path including: a first end that opens to the primary flow path at a position upstream of the impeller in a flow of air in the primary flow path, a second end that opens to the primary flow path at a position downstream of the first end, an outer circumferential wall that has a constant inner diameter along the axial direction of the secondary flow path, and an inner circumferential wall that includes a constant diameter portion that has a constant outer diameter along the axial direction of the secondary flow path and a tapered portion upstream of the constant diameter portion that is tapered toward the second end, wherein the first end opens toward a direction that is opposite to the impeller in the axial direction, the secondary flow path includes a throttle that is continuous with the first end and that is narrowed toward the first end, and the throttle is defined between the outer circumferential wall and the tapered portion of the inner circumferential wall.
 2. The centrifugal compressor according to claim 1, wherein the secondary flow path includes a plurality of fins arranged along a circumferential direction of the impeller.
 3. The centrifugal compressor according to claim 2, wherein the plurality of fins are provided in the throttle of the secondary flow path.
 4. The centrifugal compressor according to claim 2, wherein each of the plurality of fins is tilted with respect to a central axis of the impeller when seen in a radial direction of the impeller.
 5. The centrifugal compressor according to claim 4, wherein each of the plurality of fins is tilted in the same direction as leading edges of blades of the impeller when seen in the radial direction.
-